but the quality of *epistemic intervention strategies* deployed over time. Models trained with FAR, FPP, GRFR, and FTR can thus be compared not only by task success, but by the structure, timing, and efficiency of the frictive behaviors they induce.

9 Conclusion and Discussion

This paper has argued for a reframing of alignment in large language models. Rather than treating alignment as the problem of selecting the most preferred response, we have proposed to treat it as the problem of *epistemic control under uncertainty*. From this perspective, hesitation, clarification, challenge, redirection, and refusal are not failures of helpfulness, but rational control actions that regulate commitment and manage risk.

We introduced *Frictive Policy Optimization (FPO)* as a general framework for learning such control policies. The framework is grounded in three core ideas: (i) a taxonomy of frictive interventions as an explicit action space, (ii) a risk-sensitive control model of dialogue, and (iii) a structured friction functional that serves as a surrogate for epistemic and normative risk. Together, these components yield a unified family of learning methods that extend existing alignment paradigms without replacing them. Our main contributions are:

- A compact taxonomy of frictive interventions, defining a principled action space for epistemically aligned agents.
- A structured friction functional operationalizing multiple epistemic and normative failure modes, including uncertainty miscalibration, contradiction, hazard, value conflict, and information gain.
- A unified family of FPO methods, including:
  - Friction-Augmented Rewards (FAR);
  - Friction Preference Pairing (FPP);
  - Group-Relative Frictive Ranking (GRFR);
  - Friction-Conditioned Trust Regions (FTR).

  providing complementary reward-based, preference-based, ranking-based, and constraint-based mechanisms for learning epistemic intervention policies.
- An evaluation framework that measures epistemic competence directly through clarification behavior, calibration, contradiction repair, refusal proportionality, and information efficiency.

Together, these contributions provide a formal and algorithmic foundation for learning agents that are aligned not only in outcome, but in epistemic conduct.

The framework developed here has several implications for the design of aligned language models. Most importantly, it suggests that alignment cannot be achieved solely by improving the quality of answers. Instead, aligned models must learn when *not* to answer, when to seek additional information, and when to resist user requests.

This perspective casts new light on several persistent problems in alignment research. Overconfidence, hallucination, unsafe compliance, and brittle multi-turn behavior are not merely modeling errors; they are symptoms of objectives that lack a representation of intervention. By introducing friction as a first-class object of optimization, FPO offers a principled way to address these failure modes.

More broadly, the framework suggests that reflective alignment is inherently interactive. Epistemic competence is not a static property of a model's outputs, but a dynamic property of how the model manages uncertainty over time.

This work has several limitations. First, the friction functional introduced here is only a surrogate for true epistemic and normative risk. Its components rely on auxiliary models and heuristics that may themselves be imperfect and subject to Goodhart's Law: optimizing against a surrogate friction functional may induce policies that game the metric without reducing true epistemic risk. Understanding how sensitive FPO methods are to misspecification of $F(h, y)$ is an important open question. Second, we have not presented large-scale empirical results. Our focus has been on conceptual and methodological foundations. Demonstrating the practical benefits of FPO at scale remains future work. Third, the framework assumes that intervention types can be reliably annotated or inferred. In practice, distinguishing productive from unproductive friction may require careful dataset design and human judgment. Finally, we have focused primarily on dialogue. Extending FPO to tool use, planning, and long-horizon agentic behavior raises additional challenges.

We conclude by highlighting several directions for future research. A natural next step is to implement FPO methods in large-scale alignment pipelines and evaluate them on interactive benchmarks explicitly designed to elicit epistemic interventions such as clarification, deferral, and refusal. Such empirical validation would allow us to test whether policy factorization and frictional reward shaping lead to more calibrated and context-sensitive model behavior in practice.

Rather than hand-designing the friction functional, $F(h, y)$, an important extension is to learn it from human judgments of epistemic quality, yielding a second-order alignment problem in which models learn not only how to respond, but also how to internalize socially grounded norms of belief management. Extending FPO to long-horizon settings with tool use, memory, and planning would further connect this framework to recent work on agentic language models, where the timing and type of intervention are often as important as the content of the response itself.

Another promising direction is the incorporation of formal representations of ethical and legal norms into the friction functional, particularly for safety-critical or regulated domains. More broadly, FPO suggests a shift in how aligned AI systems are conceptualized: not merely as compliant assistants that optimize for immediate helpfulness, but as collaborative epistemic partners that manage uncertainty, risk, and normative constraints over time.

Alignment is often framed as the problem of making models say the right thing. We have argued that it is more fundamentally the problem of making models do the right thing at the right time. Frictive Policy Optimization offers a step toward that goal by making epistemic intervention a learnable, principled, and testable component of alignment.

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